MAHIMA BAGE

Railway Signaling Design Engineer

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PROFESSIONAL SUMMARY

Results-driven Railway Signaling Design Engineer with 4 years of experience coordinating and delivering design activities for Electronic Interlocking (EI), IR-ATP/TCAS at Hitachi Rail STS. Demonstrated ability to ensure design deliverables fulfil contract technical requirements across multi-zone Indian Railway projects (SER, ER, ECOR, NFR) and Bangladesh Rail.

KEY SKILLS

Design Coordination & Delivery: Project design activity coordination, technical requirements verification, schedule monitoring, criticality resolution, design review participation.

Subsystem Interface Management: Interface identification, definition, documentation and management across signaling and infrastructure domains.

Signaling Systems: Electronic Interlocking (EI), Microlok II, Fixed block signaling, Automatic block, ETCS Level 1 & 2, IR-ATP, Track Circuits, Axle Counters, UFSB, block instruments, LCP

Design Tools: AutoCAD, Ladder Logic Programming, Total Integration Automation (Siemens), SCADA.

Industrial Automation: PLC & SCADA

Communication Protocols: RS-232, RS-485, Ethernet, Modbus, Warm & Hot Standby Centralized & Distributed I/O

Safety & Compliance: EN 50126, EN 50128, EN 50129, CENELEC EN 50159, RAMS, SIL4, RDSO Guidelines

Documentation: Design reports, safety cases, I/O lists, interface control documents, technical standards and safety Authority.

PROFESSIONAL EXPERIENCE

Associate Signaling Design Engineer *January 2022 – Present*

Hitachi Rail STS

Coordinated and delivered Electronic Interlocking (EI) design activities across 5 Indian Railway zones (SER, ER, ECOR, NFR & WR) and Bangladesh Rail, ensuring all design outputs fulfilled contract technical requirements and complied with EN 50126, EN 50128, EN 50129, CENELEC EN 50159, and SIL4 standards.

Monitored project design execution against approved schedules, identified and resolved technical criticalities including new designs, alteration work, warm/hot standby configuration issues and Microlok model upgrade integration, maintaining on-time delivery across concurrent projects.

Led subsystem interface identification, definition, documentation, and management between Electronic Interlocking and IR-ATP/TCAS, prepared I/O lists, interface specifications, and communication protocol documents (RS-232, RS-485, Modbus) agreed across multidisciplinary teams.

Produced AutoCAD indoor interface schematics, hardware wiring diagrams, Microlok II panel designs, system load calculations, contributed to design reviews ensuring zero non-conformances in delivered design packages.

KEY PROJECTS

Indian Railway Electronic Interlocking

Contributed to EI design coordination and subsystem interface management across SER, ER, WR, ECOR, NFR, and Bangladesh Rail zones. Managed interface documentation between EI and IR-ATP/TCAS systems; supported new product development and implementation of updated Microlok Vipro into live EI systems. Ensured all deliverables met RDSO guidelines and international safety standards within agreed project schedules.

EDUCATION

PG Diploma in Railway Signaling and Telecommunication 2021

B.Tech in Electrical Engineering 2020

Maulana Abul Kalam Azad University of Technology (MAKAUT), West Bengal

Higher Secondary – ISC, St. Paul's Boarding and Day School 2016

Secondary – ICSE, St. Mary's Convent School 2014

TRAINING & LANGUAGES

Training: ETCS training from Informa Connect, Indian Railway (Kharagpur), Metro Railway Kolkata, Industrial Automation – AIAT/IIATAC Salt Lake

Languages: English